

Foundations

Abstract Algebra - Mathematics

1. Purpose of the module

This week fixes the vocabulary the rest of the course depends on: the objects being studied, the operations defined on them, and the notation used throughout. Later material assumes these definitions without restating them.

2. Definitions and notation

Terms introduced here are used in their precise technical sense, which often differs from everyday usage. Where a symbol is introduced, its meaning holds for the remainder of the module unless explicitly redefined.

3. Prerequisites

The material assumes fluency with the preceding year of study. Where a prerequisite result is needed it is stated without proof and referenced, so gaps can be filled independently.